```
xn = input("Enter the input sequence = ")
```

```
xn = 1×8
     1     2     3     4     5     6     7     8
```

```
ln = length(xn)
```

```
ln = 8
```

```
xk = zeros(1,ln)
```

```
xk = 1×8
     0     0     0     0     0     0     0     0
```

```
ixk = zeros(1,ln)
```

```
ixk = 1×8
     0     0     0     0     0     0     0     0
```

```
% obtain DFT
for k=0:ln-1
    for n=0:ln-1
        xk(k+1)=xk(k+1)+(xn(n+1)*exp((-i)*2*pi*k*n))
    end
end
```

```
xk = 1×8
     1     0     0     0     0     0     0     0
xk = 1×8
     3     0     0     0     0     0     0     0
xk = 1×8
     6     0     0     0     0     0     0     0
xk = 1×8
    10     0     0     0     0     0     0     0
xk = 1×8
    15     0     0     0     0     0     0     0
xk = 1×8
    21     0     0     0     0     0     0     0
xk = 1×8
    28     0     0     0     0     0     0     0
xk = 1×8
    36     0     0     0     0     0     0     0
xk = 1×8
    36     1     0     0     0     0     0     0
xk = 1×8 complex
  36.0000 + 0.0000i   3.0000 + 0.0000i   0.0000 + 0.0000i   0.0000 + 0.0000i ···
xk = 1×8 complex
  36.0000 + 0.0000i   6.0000 + 0.0000i   0.0000 + 0.0000i   0.0000 + 0.0000i ···
xk = 1×8 complex
  36.0000 + 0.0000i  10.0000 + 0.0000i   0.0000 + 0.0000i   0.0000 + 0.0000i ···
xk = 1×8 complex
  36.0000 + 0.0000i  15.0000 + 0.0000i   0.0000 + 0.0000i   0.0000 + 0.0000i ···
xk = 1×8 complex
  36.0000 + 0.0000i  21.0000 + 0.0000i   0.0000 + 0.0000i   0.0000 + 0.0000i ···
xk = 1×8 complex
  36.0000 + 0.0000i  28.0000 + 0.0000i   0.0000 + 0.0000i   0.0000 + 0.0000i ···
xk = 1×8 complex
  36.0000 + 0.0000i  36.0000 + 0.0000i   0.0000 + 0.0000i   0.0000 + 0.0000i ···
xk = 1×8 complex
  36.0000 + 0.0000i  36.0000 + 0.0000i   1.0000 + 0.0000i   0.0000 + 0.0000i ···
```

```
xk = 1×8 complex
  36.0000 + 0.0000i  36.0000 + 0.0000i   3.0000 + 0.0000i   0.0000 + 0.0000i ···
xk = 1×8 complex
  36.0000 + 0.0000i  36.0000 + 0.0000i   6.0000 + 0.0000i   0.0000 + 0.0000i ···
xk = 1×8 complex
  36.0000 + 0.0000i  36.0000 + 0.0000i  10.0000 + 0.0000i   0.0000 + 0.0000i ···
xk = 1×8 complex
  36.0000 + 0.0000i  36.0000 + 0.0000i  15.0000 + 0.0000i   0.0000 + 0.0000i ···
xk = 1×8 complex
  36.0000 + 0.0000i  36.0000 + 0.0000i  21.0000 + 0.0000i   0.0000 + 0.0000i ···
xk = 1×8 complex
  36.0000 + 0.0000i  36.0000 + 0.0000i  28.0000 + 0.0000i   0.0000 + 0.0000i ···
xk = 1×8 complex
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i   0.0000 + 0.0000i ···
xk = 1×8 complex
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i   1.0000 + 0.0000i ···
xk = 1×8 complex
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i   3.0000 + 0.0000i ···
xk = 1×8 complex
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i   6.0000 + 0.0000i ···
xk = 1×8 complex
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i  10.0000 + 0.0000i ···
xk = 1×8 complex
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i  15.0000 + 0.0000i ···
xk = 1×8 complex
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i  21.0000 + 0.0000i ···
xk = 1×8 complex
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i  28.0000 + 0.0000i ···
xk = 1×8 complex
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i ···
xk = 1×8 complex
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i ···
xk = 1×8 complex
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i ···
xk = 1×8 complex
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i ···
xk = 1×8 complex
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i ···
xk = 1×8 complex
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i ···
xk = 1×8 complex
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i ···
xk = 1×8 complex
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i ···
xk = 1×8 complex
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i ···
xk = 1×8 complex
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i ···
xk = 1×8 complex
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i ···
xk = 1×8 complex
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i ···
xk = 1×8 complex
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i ···
xk = 1×8 complex
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i ···
xk = 1×8
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0
xk = 1×8
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0
```

```
xk = 1×8
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0
xk = 1×8
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0
xk = 1×8
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0
xk = 1×8
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0
xk = 1×8
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0
xk = 1×8
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0
xk = 1×8
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0
xk = 1×8
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0
xk = 1×8
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0
xk = 1×8
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0
xk = 1×8
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0
xk = 1×8
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0
xk = 1×8
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0
xk = 1×8
  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0000i  36.0000 + 0.0
```

```
t=0:ln-1
```

```
t = 1×8
     0     1     2     3     4     5     6     7
```

```
subplot(221)
stem(t,xn)
ylabel("Amplitude")
xlabel("Time index")
title('Input Sequence')

% Magnitude plot
magnitude=abs(xk)
```

```
magnitude = 1×8
    36    36    36    36    36    36    36    36
```

```
t=0:ln-1
```

```
t = 1×8
     0     1     2     3     4     5     6     7
```

```
subplot(222)
stem(t,magnitude)
ylabel("Amplitude")
xlabel("k")
title("Magnitude Responce")

% phase response
```

```
phase=angle(xk)
```

```
phase = 1×8
    1.0e+-13 *
         0    0.0114    0.0229    0.0303    0.0457    0.0966    0.0607
```

```
t=0:ln-1
```

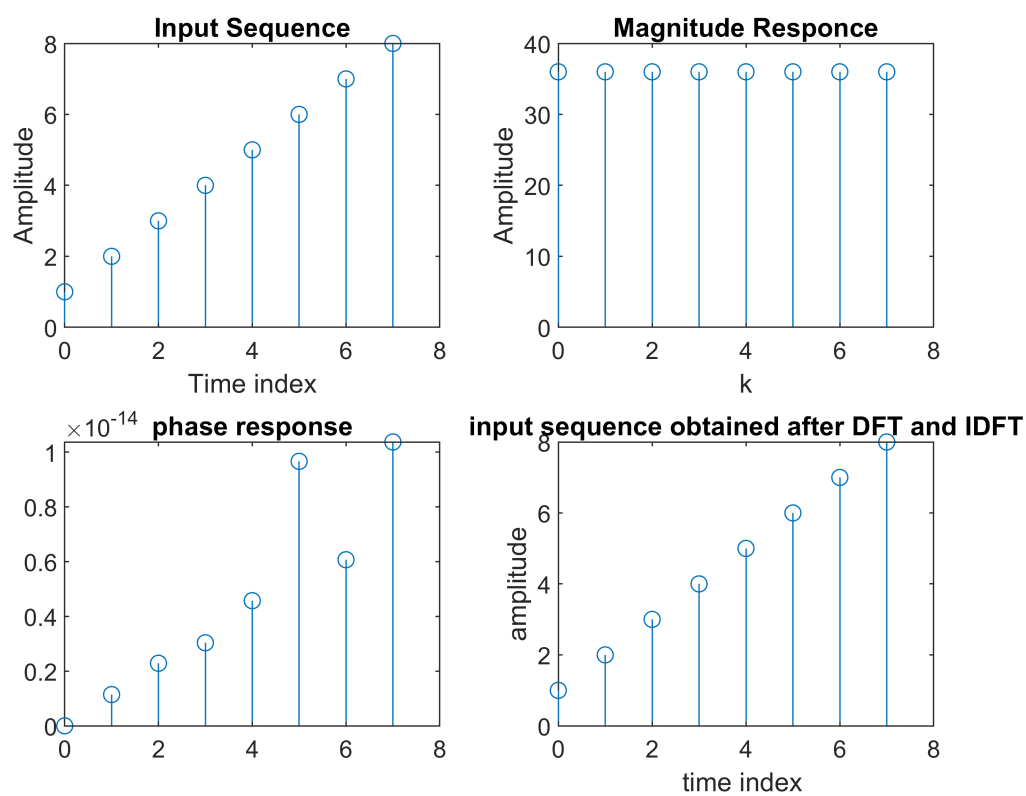
```
t = 1×8
     0     1     2     3     4     5     6     7
```

```
subplot(223)
stem(t,phase)
title('phase response')

%inverse DFT
ixk=ixk./ln
```

```
ixk = 1×8
     0     0     0     0     0     0     0     0
```

```
t=0:ln-1
```

```
t = 1×8
     0     1     2     3     4     5     6     7
```

```
subplot (224)
stem(t,xn)
xlabel("time index ")
ylabel("amplitude")
title("input sequence obtained after DFT and IDFT")
```